

#29 WhatsApp Appointment Notification Agent

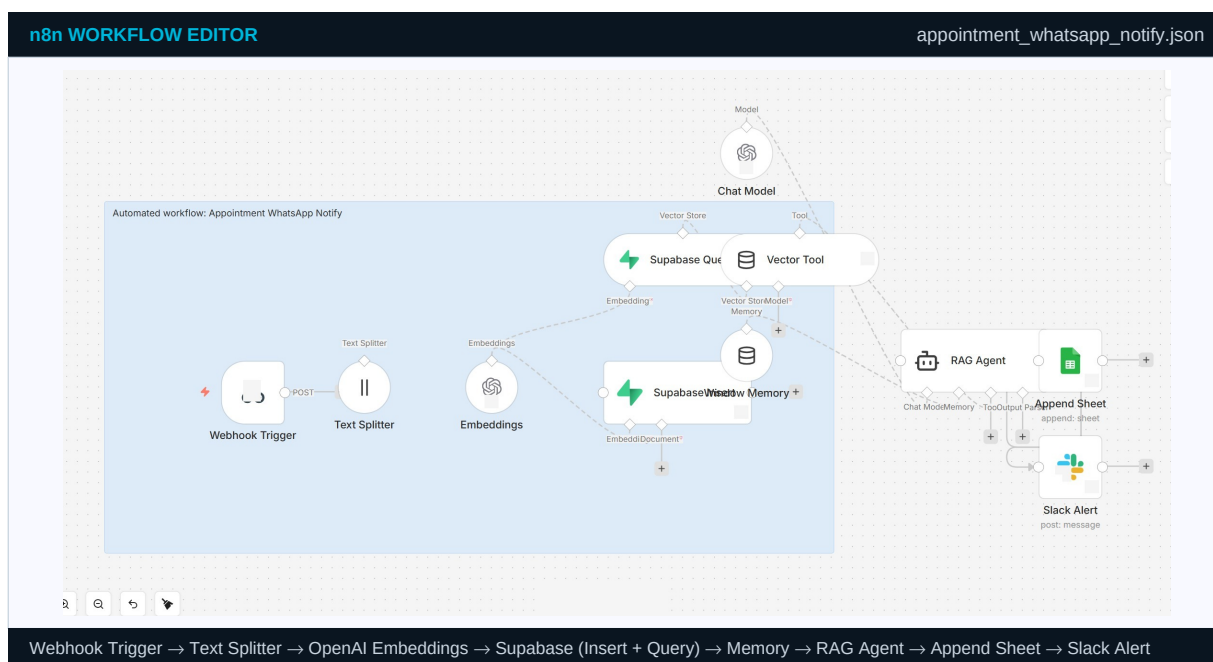
Healthcare /
Conversational
AI

Two-way WhatsApp reminders — patient confirms or reschedules automatically

BUSINESS PROBLEM

Clinics lose 15-20% of appointment slots to no-shows. Phone reminder calls take staff time and miss evening bookings. This workflow sends personalised WhatsApp reminders 24h and 2h before appointments and handles patient replies — confirm, cancel, or reschedule — autonomously updating the system.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Webhook POST — new appointment booked in HMS or Calendly

```
{
  "patient_id": "PAT-00441",
  "patient_name": "Mohammed Al-Harhi",
  "phone": "+968-9XXX-XXXX",
  "doctor": "Dr. Sarah Ahmed",
  "specialty": "Cardiology",
  "appointment": "2024-06-16T10:30:00Z",
  "location": "Building A, Floor 3, Room 312",
  "appt_type": "follow_up",
  "language": "ar"
}
```

OUTPUT — WhatsApp message sent + Sheets confirmation log

```
{
  "json": {
    "patient_id": "PAT-00441",
    "sent_at": "2024-06-15T10:30:11Z",
    "msg_status": "delivered",
    "read_at": "2024-06-15T10:47:22Z",
    "patient_reply": "CONFIRM",
    "hms_updated": true,
    "appt_status": "confirmed",
    "reminder_2h": true
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Webhook — new appointment booked	HMS or Calendly fires: {patient_id, name, phone, doctor, specialty, datetime, location, appt_type, language}.
2	✂ Splitter — clinic information chunks	Clinic FAQ, preparation instructions by appointment type, location and parking chunked for RAG.
3	■ OpenAI Embeddings — clinic knowledge	'Blood test: 8hr fast. MRI: remove all metal. Cardiology follow-up: bring previous ECG reports.'
4	■ Supabase Insert — store appointment	Full record stored: all fields plus reminder_status:PENDING, reminders_sent:0.
5	■ Supabase Query — patient history	Past appointments, no-show history, language preference, best contact time retrieved.
6	■ Vector Tool — preparation instructions	Agent fetches specific preparation for this appointment type: surgery vs scan vs routine.
7	■ Window Buffer Memory — conversation	Maintains context so patient replying RESCHEDULE triggers agent to remember which appointment.
8	■ RAG Agent — generate personalised message	Bilingual message (Arabic/English per patient preference): doctor, time, location, prep, reply options.
9	■ RAG Agent — send and handle replies	WhatsApp Business API sends message. CONFIRM/CANCEL/RESCHEDULE reply triggers HMS webhook update.
10	■ Append Sheet — appointment tracking	Logs: sent_at, delivered_at, read_at, reply, final_status. No-show KPI calculated weekly per doctor.
11	■ Slack — no-response alert at T-4h	No reply by 4 hours before appointment → Slack alert to reception for manual follow-up call.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
WhatsApp Business API Token	developers.facebook.com → WhatsApp → Business Account → Get permanent token	WhatsApp Business API
OpenAI API Key	platform.openai.com	OpenAI API
Supabase URL + Anon Key	app.supabase.com → Settings → API	Supabase API
Slack Bot Token	api.slack.com/apps → OAuth & Permissions	Slack API

SAMPLE RUN — what this workflow processed and produced

Scenario	A private cardiology clinic runs 45-60 appointments per day. No-show rate was 18%. Phone reminders took a nurse 90 minutes each morning.
Input received	New appointment: Mohammed Al-Harhi, Cardiology follow-up, Dr. Sarah Ahmed, 16 June 10:30, Room 312. Patient language preference: Arabic.

What the workflow found	Patient had one previous no-show (March 2024). Preparation: 'bring previous ECG reports.' Arabic message template selected.
Output produced	WhatsApp reminder sent at 10:30 (24h before). Read at 10:47. Patient replied CONFIRM at 10:47. HMS updated automatically. No-show rate for WhatsApp-reminded patients: 6% vs 18% phone-only group.

JSON: appointment_whatsapp_notify.json	GitHub: trm-healthcare-whatsapp-agent	business.whatsapp.com/
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